Introduction to Python Programming

Python is a general-purpose interpreted, interactive, object-oriented, and high-level programming language. Python is dynamically-typed and garbage-collected programming language. It was created by Guido van Rossum.

Python has two basic modes: script and interactive. The normal mode is the mode where the scripted and finished .py files are run in the Python interpreter. Interactive mode is a command line shell which gives immediate feedback for each statement, while running previously fed statements in active memory. As new lines are fed into the interpreter, the fed program is evaluated both in part and in whole. Interactive mode is a good way to play around and try variations on syntax.

Features:
1) Easy to Learn and Use - Python is easy to learn and use.

It is developer-friendly and high level programming language.
2) Expressive Language - Python language is more expressive means that it is more understandable and readable.
3) Interpreted Language - Python is an interpreted language i.e. interpreter executes the code line by line at a time. This makes debugging easy and thus suitable for beginners.
4) Cross-platform Language - Python can run equally on different platforms such as Windows, Linux, Unix and Macintosh etc. So, we can say that Python is a portable language.
5) Free and Open Source - Python language is freely available at offical web address.The source-code is also available. Therefore it is open source.
6) Object-Oriented Language - Python supports object oriented language and concepts of classes and objects come into existence.
7) Extensible - It implies that other languages such as C/C++ can be used to compile the code and thus it can be used further in our python code.
8) Large Standard Library - Python has a large and broad library and prvides rich set of module and functions for rapid application development.
9) GUI Programming Support - Graphical user interfaces can be developed using Python.
10) Integrated - It can be easily integrated with languages like C, C++, JAVA etc.

First Python Program
Let us write a simple Python program in a script. Python files have extension .py. Type the following source code in a test.py file −

```
print "Hello, Python!"
```

Reserved Keywords: 31 reserved keywords are there

Lines and Indentation
Python provides no braces to indicate blocks of code for class and function definitions or flow control. Blocks of code are denoted by line indentation, which is rigidly enforced.
The number of spaces in the indentation is variable, but all statements within the block must be indented the same amount. For example −

```
if True:
  print "True"
else:
  print "False"
```

Quotation in Python

Python accepts single ('), double (") and triple (''' or """) quotes to denote string literals, as long as the same type of quote starts and ends the string.

The triple quotes are used to span the string across multiple lines. For example, all the following are legal −

word = 'word'
sentence = "This is a sentence."
paragraph = """This is a paragraph. It is
made up of multiple lines and sentences."""

Assigning Values to Variables

```
counter = 100          # An integer assignment
miles   = 1000.0       # A floating point
name    = "John"       # A string
print counter
print miles
print name
```

Multiple Assignment

```
a = b = c = 1
a,b,c = 1,2,"john"
```

Standard Data Types

Python has five standard data types −

- Numbers
- String
- List
- Tuple
- Dictionary

Python supports four different numerical types −

- int (signed integers)
- long (long integers, they can also be represented in octal and hexadecimal)
- float (floating point real values)
- complex (complex numbers)

Problem Statement:

1. Do the following in the Python shell.
   a) Save your name and print it.
   >>>name='John'
   >>>print(name)
   John

   b) Take input your name and print it as "Hello! John", where "John" is the input.
   >>>name=input("Enter your name")
   DC
   >>>print("Hello! "+name)
   Hello! DC
   >>>print("Hello! ",name)

Hello! DC

c) Add two numbers. Input the two numbers.
```
>>>a=input("Enter a number:")
5
>>>b=input("Enter another number:")
12
>>>c=a+b
>>>print(c)
17
```

d) Subtract two numbers. Input the two numbers.
```
>>>a=input("Enter a number:")
12
>>>b=input("Enter another number:")
10
>>>c=a-b
>>>print(c)
2
```

e) Multiply two numbers. Input the two numbers.
```
>>>a=input("Enter a number:")
5
>>>b=input("Enter another number:")
12
>>>c=a*b
>>>print(c)
60
```

f) Divide two numbers. Print remainder. Input the two numbers.
```
>>>a=input("Enter a number:")
5
>>>b=input("Enter another number:")
12
>>>c=a/b
>>>print(c)
2.4
>>>print(a%b)
2
```

g) Divide two numbers truncating the fractional part. Input the two numbers.
```
>>>a=input("Enter a number:")
5
>>>b=input("Enter another number:")
12
>>>c=a//b
>>>print(c)
2
```

h) Find a^b. Input the two numbers.
```
>>>a=input("Enter a number:")
2
>>>b=input("Enter another number:")
4
```

```
>>>c=a**b
>>>print(c)
8
```

i) Print the following any strings.

```
>>>print("Yes, he said. It doesn’t matter.")
Yes, he said. It doesn’t matter.
>>>print('"Isn\'t" she said.')
"Isn't" she said.
```

j) Write a string and print 0^{th} letter and 5^{th} letter.

```
>>> st= "Python"
>>>st[0]
P
>>>st[5]
n
```

k) Write two words with an escape sequence in between. Now print it.

```
>>>st="Python\n Java"
>>>print(st)
Python
 Java
```

l) Perform integer conversion.

```
>>> int('52')
52
```

m) Perform string conversion

```
>>> str(52)
'52'
```

n) Concatenate two strings without using ‘+’.

```
>>> "hello" "good"
'hellogood'
```

o) Concatenate two strings using ‘+’.

```
>>> str1="Hello"
>>> str2="Good"
>>> str3=str1+str2
>>> print(str3)
HelloGood
```

o) Take input of a floating-point number and an integer. Multiply them. Add three with the result without saving it as a separate variable. Also, round it up to two decimal places without saving it as a separate variable.

```
>>> x=5.789
>>> y=13
>>> x*y
75.25699999999999
>>> 3+ _
78.25699999999999
>>> round(_ ,2)
78.26
```

2. Write a program which contains stores a string containing multiple lines. Print it as it is.

```
print(''' Hello! Good Morning)

Output:
How are you?
How did you spend your holidays?''')
 Hello! Good Morning
How are you?
How did you spend your holidays?
```

3. Write a program to prompt the user for hours and wages as rate per hour to compute gross pay.

```
hours=input("Enter total hours")
wage=input("Enter rate per hour")
print("The gross pay is: "+ round(hours*wage,2))
```

Questionnaires:

1. Write down the Python keywords.
2. Differentiate between script and program.
3. What are the features of Python programming language?

Assignment:

1. Write a program which takes temperature as input as Centigrade. Convert it to Fahrenheit.
2. Write a program to prompt the user for principal amount, annual interest, and years. Print total interest and total amount including principal.

4. Write a program which takes temperature as input as Centigrade. Convert it to Fahrenheit.

```
#Python Program to convert temperature in celsius to fahrenheit
# change this value for a different result
celsius = 37.5
# calculate fahrenheit
fahrenheit = (celsius * 1.8) + 32
print('%0.1f degree Celsius is equal to %0.1f degree Fahrenheit' %(celsius,fahrenheit))
```

5. Write a program to prompt the user for principal amount, annual interest, and years. Print total interest and total amount including principal.

```
principal=input("Enter principal")
interest=input("Enter Annual Interest")
years=input("Enter Number of Years")
Total=principal*years*interest/100
print("The total interest is:" + Total)
Gross=Total+principal
print("The total amount is:" + Gross)
```